BOTANICAL SOURCE





Zanthoxylum chalybeum

15 natural products

LIGAND PREPARATION
Protonation at pH 5.0

RECEPTOR PREPARATION
Plasmepsin II (PMII)
Protonation at pH 5.0

MOLECULAR DOCKING
AutoDock Vina
15 natural products

DOCKING VALIDATION
R37 redocking

TOP DOCKING HITS
Docking score (kcal/mol)
THL: **–8.411**
SES: **–8.369**
NCH: **–8.138**

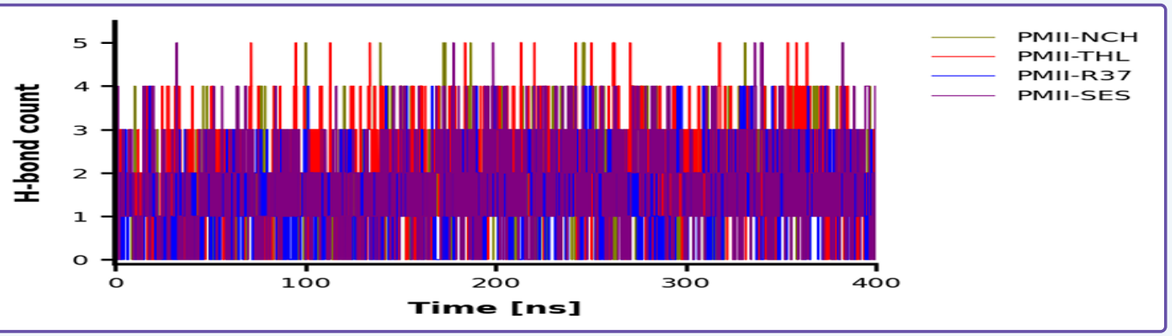
MOLECULAR DYNAMICS
400 ns per system
Apo PMII | R37 | THL
SES | NCH

TRAJECTORY ANALYSIS
RMSD, RMSF, R_g, SASA
Binding-site retention
Hydrogen-bond dynamics

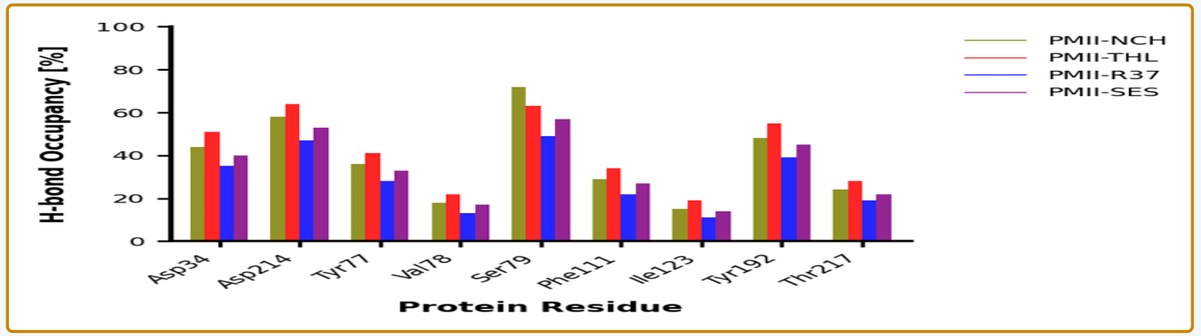
MM/PBSA ANALYSIS
Final 200 ns (200–400 ns)
200 snapshots per complex
Binding-energy estimation

LEAD CANDIDATE: 3,20,28-TRIHYDROXYLUPANE (THL)

H-BOND COUNT DYNAMICS (400 ns)



RESIDUE-SPECIFIC H-BOND OCCUPANCY



Docking score: **–8.411** kcal/mol • Contact retention: **82.8%** • Mean H-bonds: **1.93**
Asp34 occupancy: **51%** • Asp214 occupancy: **64%** • MM/PBSA: **–47.38** ± **2.72** kcal/mol